1 **(i)** Find the expansion of $(8-x)^{\frac{1}{3}}$ in ascending powers of x up to and including the term in x^2. [2]

(ii) By substituting $x=-1$, find an approximation of $\sqrt[3]{3}$, leaving your answer in fraction. [2]

(iii) The substitution $x=5$ can also be used to find an approximation for $\sqrt[3]{3}$. Explain whether this will give a better approximation than using $x=-1$. [1]

2 **(i)** Using an algebraic method, solve the inequality $\dfrac{2x+11}{x^2+2}>4$, leaving your answer in exact form. [3]

(ii) Hence by completing the square or otherwise, solve the inequality

$$\frac{2x+7}{x^2-4x+6}>4.$$ [3]

3 Given that $y=e^{3\tan^{-1}x}$, show that

$$\left(1+x^2\right)\frac{dy}{dx}=3y.$$

By repeated differentiation of this result, find the Maclaurin's expansion for y, up to and including the term in x^3. [5]

Hence, deduce the Maclaurin's expansion for $e^{2x+3\tan^{-1}x}$, up to and including the term in x^2. [2]

4 The first $4n$ terms of a series are obtained by adding the first $2n$ terms of two geometric progressions. The first geometric progression has first term 2 and the second geometric progression has first term $2b$. The common ratios of the two progressions are both equal to $-b^2$ where $b^2 \neq -1$.

(i) Find the sum of the first $4n$ terms of the series, simplifying your answer in terms of b. [2]

(ii) State the range of values of b for which the series has a sum to infinity.
If $b=-0.9$, find the smallest value of n such that the sum to infinity of the series is more than the sum of the first $4n$ terms by less than 0.001. [4]

5 On a clearly labelled Argand diagram, sketch the locus $|z+2-2\mathrm{i}|=4$.

Sketch, on the same diagram, the locus $\arg(z+2+4\mathrm{i})=\theta$, where $0<\theta<\frac{\pi}{2}$, such that there is only one point of intersection. Show that $\sin\theta=\frac{\sqrt{5}}{3}$.

Hence find w, the complex number representing the intersection point of the two loci in the form $x+\mathrm{i}y$, giving the exact values of x and y. [6]

6 Two planes π_1 and π_2 have a common point $\boldsymbol{A}$ with position vector given by $5\mathbf{i}-3\mathbf{j}$. These two planes π_1 and π_2 contain ℓ_1 and ℓ_2 respectively, with vector equations given by:

$$\ell_1: \quad \mathbf{r}=(-\mathbf{j}+\mathbf{k})+\lambda(2\mathbf{i}+\mathbf{j}-\mathbf{k}), \text{ where } \lambda\in\mathbb{R},$$

$$\ell_2: \quad \mathbf{r}=(\mathbf{i}-\mathbf{j}+\mathbf{k})+\mu(\mathbf{j}-\mathbf{k}), \text{ where } \mu\in\mathbb{R}.$$

(i) Find the vector equation of the line of intersection ℓ_3 of the two planes π_1 and π_2. [5]

(ii) Hence find the acute angle between ℓ_3 and the plane π_3 which contains both lines ℓ_1 and ℓ_2. [3]

7 The parametric equations of a curve C are

$$x=e^t\sin t \quad \text{and} \quad y=e^t\cos t \quad \text{where } -\pi\le t\le\pi.$$

(i) Sketch the curve C, showing clearly the axial intercepts. [2]

(ii) Find the equation of the **tangent** to the curve at the point P where $t=\frac{\pi}{4}$. [3]

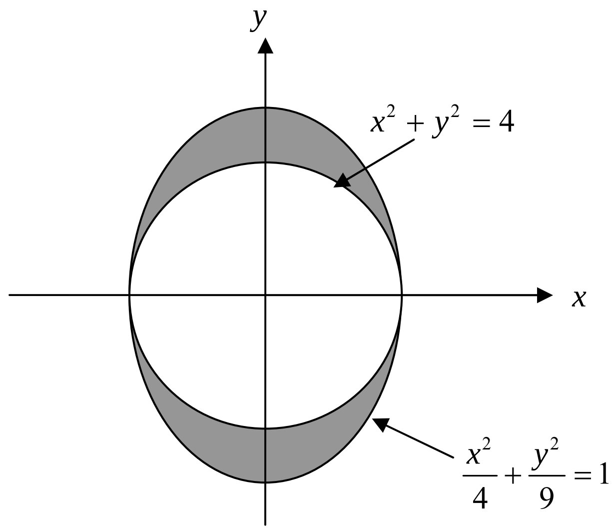
(iii) Given that O is the origin and Q is the point on the curve C such that $\angle POQ=\frac{\pi}{12}$, where the angle OQ made with the positive x-axis is **bigger** than the angle OP made with the positive x-axis. Find the area of the triangle POQ. [3]

8(a) Show that $\dfrac{2x^2-5x+13}{x^2-2x+5}$ can be written in the form $A+\dfrac{B(2x-2)+C}{x^2-2x+5}$ where A, B and C are constants to be determined.

Hence find the integral $\displaystyle\int \frac{2x^2-5x+13}{x^2-2x+5}\,dx$. [5]

(b) Find, in terms of n and e, $\displaystyle\int_1^{2e} x^{n-1}\ln x\,dx$. [4]

9



The diagram above shows the shaded region R enclosed by the curves $x^2+y^2=4$ and $\dfrac{x^2}{4}+\dfrac{y^2}{9}=1$.

(a) Using the substitution $x=2\sin\theta$, find the exact area of region R. [5]

(b) Find the exact volume of the solid formed when R is rotated through π radians about the y-axis. [3]

10 The curve C has equation $y=\dfrac{2x^2+3x-15}{x(x+\lambda)}$, where λ is a constant.

(i) Write down the equations of the asymptotes of C and the intercepts with the x-axis. [3]

(ii) If $\lambda = 2$, find the coordinates of the turning points of the curve C, expressing your answer in three significant figures.
Sketch the curve C for $\lambda = 2$, indicating on your diagram, the turning points, asymptotes and intercepts with the x-axis. [4]

(iii) Sketch the curve C, on a separate diagram, for the case where $\lambda = 4$. Show clearly the asymptotes and the intercepts with the x-axis. [2]

Hence, by drawing a sketch of another suitable curve in the diagram of **(iii)**, show that the equation $x^4+4x^3-2x^2-3x+15=0$ has exactly two real roots. [2]

11(a) **(i)** Solve the equation $z^4 = -64$, giving your answers in exponential form. [4]

(ii) By writing your answers to part **(i)** in the form $x + \mathrm{i}y$, find the four linear factors of $z^4 + 64$. [2]

(iii) Hence express $z^4 + 64$ as a product of 2 real quadratic factors. [1]

(b) Given that $e^{\mathrm{i}\theta}=\cos\theta+\mathrm{i}\sin\theta$ and $e^{-\mathrm{i}\theta}=\cos\theta-\mathrm{i}\sin\theta$.

By expressing $\cos\theta$ and $\sin\theta$ in terms of $e^{\mathrm{i}\theta}$ and $e^{-\mathrm{i}\theta}$, show that

$$\cos^2\theta\sin^4\theta=\frac{1}{32}(\cos 6\theta-2\cos 4\theta-\cos 2\theta+2).$$

Hence without the use of a calculator, find the exact value of $\int_0^{\frac{\pi}{3}}\cos^2\theta\sin^4\theta\,\mathrm{d}\theta$.

[6]

12(a) The sum of the first n terms of a series is given by $S_n = n^2 + \frac{1}{3^n}$.

(i) Find the n^{th} term of the series, T_n. [2]

(ii) By using the formula for T_n and S_n, evaluate

$$7 - \frac{2}{3^4} + 9 - \frac{2}{3^5} + 11 - \frac{2}{3^6} + 13 - \frac{2}{3^7} + 15 - \frac{2}{3^8},$$

leaving your answer in improper fraction. [2]

(iii) Without the use of a calculator, estimate $\sqrt{S_{10^{20}}}$ to the nearest integer. [2]

(b) The interior of a rectangle is divided into regions by lines where no two lines are parallel and no three lines pass through the same point. The number of regions formed by $n + 1$ such lines is given by the relation

$$x_{n+1} = n + 1 + x_n, \quad x_0 = 1$$

(i) Find the values of x_1, x_2, x_3 and x_4. [2]

(ii) A conjecture for x_n in terms of n is given by $x_n = S_n + 1$ where S_n is the sum of the first n terms of a certain arithmetic progression. Find the conjecture for x_n in terms of n. [2]

(iii) Hence, by using mathematical induction, prove the conjecture for all non- negative integer values of n. [3]

The End